Web Scraping using Python

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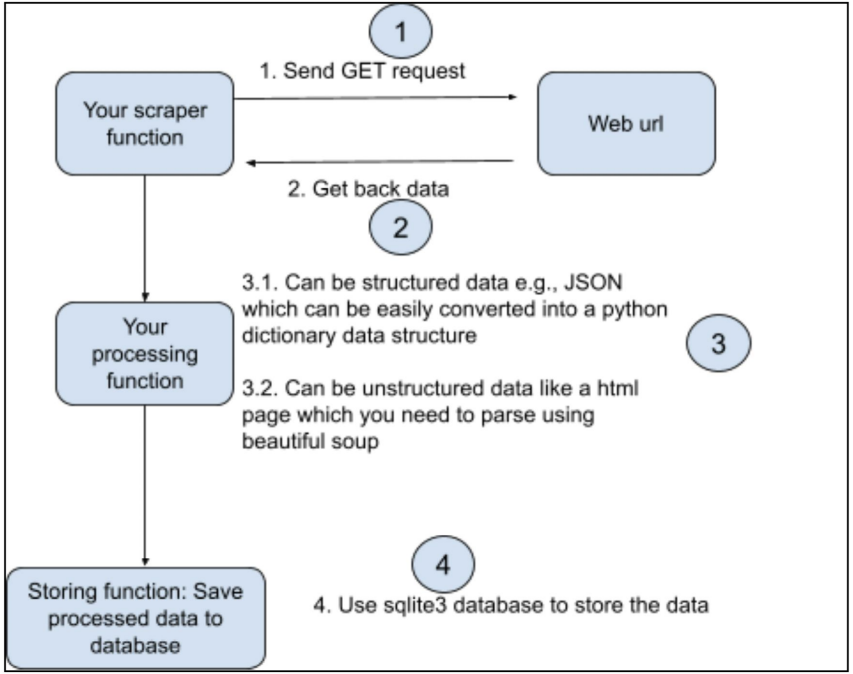
What is Web Scrapping

Web scrapers are programs that can collect data from websites, parse it and store the data (e.g., for analysis later).

A web request can be GET or POST. Using GET requests you can get data from the world wide web (WWW).

Pipeline for Scraping

1. Send the request (GET/POST)
2. Get the response
3. Based on response type; process the data
4. Store the data



Tools we will be learning

1. **Urllib:** “A built-in Python library used to fetch URLs and handle low-level HTTP requests, responses, headers, and cookies.”
2. **Requests**: “allows you to send HTTP/1.1 requests extremely easily”
3. **BeautifulSoup (BS4):** “scrape information from web pages. Sits atop an HTML parser, assists iterating, searching, and modifying the parse tree.”
4. **Selenium**: “Primarily it is for automating web applications for testing purposes, but is certainly not limited to just that.”
5. **Sqlite3**: “provides a lightweight disk-based database that doesn’t require a separate server process and allows accessing the database using a nonstandard variant of the SQL query language.”

What is urllib?

1. urllib is a standard Python library for:
    a. Fetching URLs
    b. Sending HTTP/HTTPS requests
    c. Handling responses
2. Does not require installation
3. Lower-level than requests, but more transparent

Main submodules:

- **urllib.request** – open URLs
- **urllib.parse** – URL encoding/decoding
- **urllib.error** – error handling

Urllib - parse

```
from urllib.parse import urlparse
urlparse("scheme://netloc/path;parameters?query#fragment")
o =
urlparse("http://docs.python.org:80/3/library/urllib.parse.html?"
             "highlight=params#url-parsing")
print(o)
print(o.scheme)
print(o.netloc)
print(o.hostname)
print(o.port)
print(o._replace(fragment="").geturl())
```

```
ParseResult(scheme='http', netloc='docs.python.org:80', path='/3/library/urllib.parse.html', params='',
query='highlight=params', fragment='url-parsing')
http
docs.python.org:80
docs.python.org
80
http://docs.python.org:80/3/library/urllib.parse.html?highlight=params
```

Urllib - request

```
import urllib.request
req = urllib.request.Request('http://www.example.com/')
req.add_header('Referer', 'http://www.python.org/')
# Customize the default User-Agent header value:
req.add_header('User-Agent', 'urllib-example/0.1 (Contact: . .
.)')
with urllib.request.urlopen(req) as f:
    print(f.read().decode('utf-8'))
```

```
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport"
content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;
font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</
style><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without
needing permission. Avoid use in operations.<p><a href="https://iana.org/domains/example">Learn more</
a></div></body></html>
```

Requests Library

1. requests is a high-level HTTP library in Python
2. Simplifies sending HTTP requests compared to urllib
3. Handles headers, cookies, sessions, and authentication easily

Key features:

- Simple and readable syntax
- Automatic JSON handling
- Built-in cookie & session management
- Widely used in industry and research

Requests Library - Example

1. **Sending a GET request:**

```
response = requests.get("https://httpbin.org/get")
print(response.text)
```

2. **Sending headers with requests:**

```
headers = {"User-Agent": "Mozilla/5.0",
    "Accept": "application/json"}
response = requests.get("https://httpbin.org/headers", headers=headers)
print(response.json())
```

BeautifulSoup

```
html_doc = """
<html><head><title>The Dormouse's story</title></head>
<body>
<p class="title"><b>The Dormouse's story</b></p>

<p class="story">Once upon a time there were three little sisters; and their names were
<a href="http://example.com/elsie" class="sister" id="link1">Elsie</a>,
<a href="http://example.com/lacie" class="sister" id="link2">Lacie</a> and
<a href="http://example.com/tillie" class="sister" id="link3">Tillie</a>;
and they lived at the bottom of a well.</p>

<p class="story">...</p>
"""
```

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(html_doc, 'html.parser')
```

Here are some simple ways to navigate that data structure:

```
soup.title
# <title>The Dormouse's story</title>

soup.title.name
# u'title'

soup.title.string
# u'The Dormouse's story'

soup.title.parent.name
# u'head'

soup.p
# <p class="title">><b>The Dormouse's story</b></p>

soup.p['class']
# u'title'

soup.a
# <a class="sister" href="http://example.com/elsie" id="link1">Elsie</a>

soup.find_all('a')
# [<a class="sister" href="http://example.com/elsie" id="link1">Elsie</a>,
#  <a class="sister" href="http://example.com/lacie" id="link2">Lacie</a>,
#  <a class="sister" href="http://example.com/tillie" id="link3">Tillie</a>]

soup.find(id="link3")
# <a class="sister" href="http://example.com/tillie" id="link3">Tillie</a>
```

BeautifulSoup: Searching the tree

Two most popular methods: find() and find_all()

You can search for an element using different kinds of filters:

- `String: soup.find_all('b')`
- `Regex : soup.find_all(re.compile("^b"))`
- `List  : soup.find_all(["a", "b"])`

find_all(name, attrs, recursive, string, limit, **kwargs)

```
soup.find_all("title")
# [<title>The Dormouse's story</title>]

soup.find_all("p", "title")
# [<p class="title"><b>The Dormouse's story</b></p>]

soup.find_all("a")
# [<a class="sister" href="http://example.com/elsie" id="link1">Elsie</a>,
#  <a class="sister" href="http://example.com/lacie" id="link2">Lacie</a>,
#  <a class="sister" href="http://example.com/tillie" id="link3">Tillie</a>]

soup.find_all(id="link2")
# [<a class="sister" href="http://example.com/lacie" id="link2">Lacie</a>]

import re
soup.find(string=re.compile("sisters"))
# u'Once upon a time there were three little sisters; and their names were\n'
```

find_all(name, attrs, recursive, string, limit, **kwargs)

1. Find tags starting with specific letters:
   a. **soup.find_all(re.compile("^h")) -** Matches: <html>, <head>, <h1>, <h2>
2. Find multiple tag types:
   a. **soup.find_all(re.compile("div|span"))**
3. Match class names starting with product:
   a. **soup.find_all("div", class_=re.compile("^product-"))**
4. Match IDs ending with numbers”
   a. **soup.find_all(id=re.compile(r"\d+$"))**
5. Find all HTTPS links
   a. **soup.find_all("a", href=re.compile("^https://"))**
6. Exclude certain links (negative regex)
   a. **soup.find_all("a", href=re.compile("^(?!.*logout)"))**

find(name, attrs, recursive, string, **kwargs)

The *find_all()* method scans the entire document looking for results, but sometimes you only want to find one result. — use **find()**

```
soup.find_all('title', limit=1)
# [<title>The Dormouse's story</title>]

soup.find('title')
# <title>The Dormouse's story</title>
```

find_all_next, find_next

These methods use iterates over whatever tags and strings that come after it in the document

```
first_link = soup.a
first_link
# <a class="sister" href="http://example.com/elsie" id="link1">Elsie</a>

first_link.find_all_next(string=True)
# [u'Elsie', u',\n', u'Lacie', u' and\n', u'Tillie',
#  u';\nand they lived at the bottom of a well.', u'\n\n', u'...', u'\n']

first_link.find_next("p")
# <p class="story">...</p>
```

Selenium: Writing your first script

- Start the session
  ```
  driver = webdriver.Chrome()
  ```
- Take action on browser
  ```
  driver.get("https://www.selenium.dev/selenium/web/web-form.html")
  ```
- Request browser information
  ```
  title = driver.title
  ```
- Find an element
  ```
  text_box = driver.find_element(by=By.NAME, value="my-text")
  submit_button = driver.find_element(by=By.CSS_SELECTOR, value="button")
  ```

Selenium: Writing your first script

- Take action on element

```
text_box.send_keys("Selenium")
submit_button.click()
```

- Request element information

```
text = message.text
```

- End the session

```
driver.quit()
```

Selenium: Finding Elements Using XPath

1. **Basic XPath selection:**

   ```
   element = driver.find_element(By.XPATH, "//input[@name='username']")
   ```

2. **XPath using contains():**

   ```
   driver.find_element(By.XPATH, "//button[contains(text(),'Submit')]")
   ```

3. **Select element by partial class name:**

   ```
   driver.find_element(By.XPATH, "//div[contains(@class,'product')]")
   ```

4. **Select nth element:**

   ```
   driver.find_element(By.XPATH, "(//input[@type='text'])[2]")
   ```

5. **Select child element:**

   ```
   driver.find_element(By.XPATH, "//div[@id='menu']/a[1]")
   ```

Selenium: Clicking and Sending Input

1. **Send input to text box:**

```
textbox = driver.find_element(By.XPATH, "//input[@id='email']")
textbox.send_keys("test@example.com")
```

2. **Click a button:**

```
button =
driver.find_element(By.XPATH,"//button[@type='submit']")
button.click()
```

Sqlite3: Writing your first script

- Create a new database and open a database connection
```
import sqlite3
con = sqlite3.connect("tutorial.db")
```
- Create a database cursor: to execute SQL statements and fetch results from SQL queries
```
cur = con.cursor()
```
- Create a database table movie with columns for title, release year, and review score
```
cur.execute("CREATE TABLE movie(title, year, score)")
```
- Add data supplied as SQL literals by executing an INSERT statement
```
cur.execute("""
     INSERT INTO movie VALUES
     ('Monty Python and the Holy Grail', 1975, 8.2)
""")
```

Sqlite3: Writing your first script

- Fetching results from a table by executing SELECT
```
res = cur.execute("SELECT score FROM movie")
res.fetchall()
```
- Saving the changes
```
con.commit()
```
- Closing the db connection
```
con.close()
```